TWO LOCALLY CLOSED PIECES THAT ARE DISJOINT OVER A QUASI-COMPACT OPEN CAN BE SEPARATED BY AN ADMISSIBLE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F3V (sources/stacks-flat/pinned-original-source.tex line 9554).

Allowed standard prerequisites (editorial).

- Finite presentation of a closed subscheme and finite type of its ideal sheaf. Morphisms, Lemma labelled morphisms-lemma-closed-immersion-finite-presentation, cited at proof lines 9568-9572. Allows the ideals I_i to be used in a blowup construction.
- Separating disjoint closed subschemes over an open by an admissible blowup. Divisors, Lemma labelled divisors-lemma-separate-disjoint-opens-by-blowing-up, cited at proof lines 9574-9579. Produces the blowup of Z making the strict transforms disjoint.
- Strict transforms under a blowup and the composition of finite presentation morphisms. Divisors, Lemma labelled divisors-lemma-strict-transform, cited at proof lines 9594-9598, and Morphisms, Lemma labelled morphisms-lemma-composition-finite-presentation, cited at proof lines 9587-9591. Shows the blowup of the center Y on X is admissible and computes the strict transforms.
- Constructible closed subsets as zero loci of finite type ideals, and extension of ideals from an open. Properties, Lemmas labelled properties-lemma-quasi-coherent-finite-type-ideals and properties-lemma-extend, cited at proof lines 9595-9603. Reduces part (2) to part (1).

Source reference. Original lemma 01TV: sources/stacks-chow/tags.txt, source lines 2388–2388. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9572 of the counted proof. It is

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Source reference. Original lemma 080P: sources/stacks-chow/tags.txt, source lines 10189–10189. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9577 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 080K: sources/stacks-chow/tags.txt, source lines 10185–10185. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9583 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01TR: sources/stacks-chow/tags.txt, source lines 2384–2384. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9584 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080E: sources/stacks-chow/tags.txt, source lines 10179–10179. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9590 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01PH: sources/stacks-chow/tags.txt, source lines 2230–2230. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9595 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01PF: sources/stacks-chow/tags.txt, source lines 2228–2228. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9599 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $U \subset X$ be a quasi-compact open.*

1. *If $Z_1, Z_2 \subset X$ are closed subschemes of finite presentation such that $Z_1 \cap Z_2 \cap U = \emptyset$, then there exists a U-admissible blowing up $X' \to X$ such that the strict transforms of Z_1 and Z_2 are disjoint.*
2. *If $T_1, T_2 \subset U$ are disjoint constructible closed subsets, then there is a U-admissible blowing up $X' \to X$ such that the closures of T_1 and T_2 are disjoint.*

Proof. Proof of (1). The assumption that $Z_i \to X$ is of finite presentation signifies that the quasi-coherent ideal sheaf $\mathcal{I}_i$ of Z_i is of finite type, see Morphisms, Lemma 01TV. Denote $Z \subset X$ the closed subscheme cut out by the product $\mathcal{I}_1\mathcal{I}_2$. Observe that $Z \cap U$ is the disjoint union of $Z_1 \cap U$ and $Z_2 \cap U$. By Divisors, Lemma 080P there is a $U \cap Z$-admissible blowup $Z' \to Z$ such that the strict transforms of Z_1 and Z_2 are disjoint. Denote $Y \subset Z$ the center of this blowing up. Then $Y \to X$ is a closed immersion of finite presentation as the composition of $Y \to Z$ and $Z \to X$ (Divisors, Definition 080K and Morphisms, Lemma 01TR). Thus the blowing up $X' \to X$ of Y is a U-admissible blowing up. By general properties

of strict transforms, the strict transform of Z_1, Z_2 with respect to $X' \to X$ is the same as the strict transform of Z_1, Z_2 with respect to $Z' \to Z$, see Divisors, Lemma 080E. Thus (1) is proved.

Proof of (2). By Properties, Lemma 01PH there exists a finite type quasi-coherent sheaf of ideals $\mathcal{J}_i \subset \mathcal{O}_U$ such that $T_i = V(\mathcal{J}_i)$ (set theoretically). By Properties, Lemma 01PF there exists a finite type quasi-coherent sheaf of ideals $\mathcal{I}_i \subset \mathcal{O}_X$ whose restriction to U is $\mathcal{J}_i$. Apply the result of part (1) to the closed subschemes $Z_i = V(\mathcal{I}_i)$ to conclude. □